

ENGG100

Projectile Simulation Project Final Report

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Problem Statement:

The problem on hand consists of the 3D projection of an object at an elevation angle of θ from the horizontal axis(x axis) from the point A which has the coordinates $(x_1, y_1, 0)$ to the point B which has the coordinates $(x_1 + D + L + 10, 5-y_1, 0)$ as we are not considering movement in the z axis the coordinate will remain as 0, where D is the difference in distance between x_1 and the building, L being the width of the building and H being the height of the building, the building will be located between both points blocking the line of sight to point B which the object will have to surpass vertically, landing on the target spot which is 5m above the ground and 10m horizontally away from the end-base of the building. The problem will be solved using motion equations relating to projectiles and implementing simple Newtonian mechanics.

Objectives:

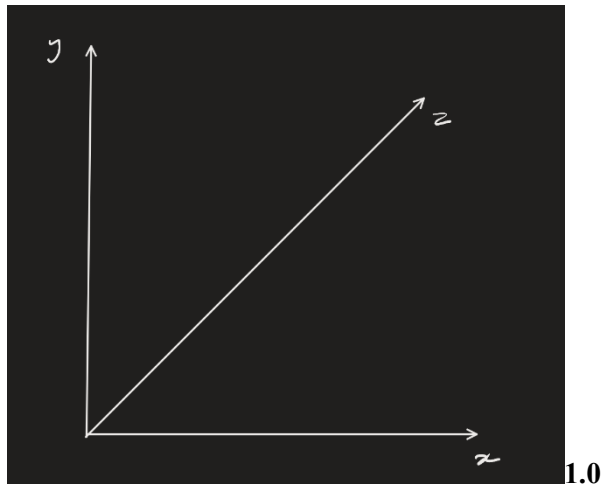
1. Analyse the projectile motion using Engineering Dynamics:
 - Use relevant equations of motion to evaluate the trajectory of the projectile launched from coordinate A.
 - Consider the effect of gravity and 3D coordinates in the analysis.
2. Determine Launch Parameters:
 - Calculate the required velocity (V) and launch angle (θ) for the projectile to land at coordinate B, while avoiding collision with the obstruction (building) located between the the launch and landing points.
3. Develop a MATLAB based simulation:
 - Build a MATLAB program that allows user to input different launch coordinates, and building height, and outputs the required launch velocity and angle.
 - Include features for graphing the projectile's path, displaying maximum height, and checking the projectile's location at any time step.
4. Create a User-Friendly application:
 - Implement a Graphical User Interface (GUI) in MATLAB to make the simulation accessible.
 - Allow toggling between GUI and non-GUI modes for visualisation and interaction.
5. Explore the effects of Variable Inputs:
 - Analyse how the projectile path, and feasibility of hitting the target are affected by the initial conditions.
 - Use simulations to validate theoretical predictions.

Theoretical Analysis and Methodology:

1. Modeling the problem:

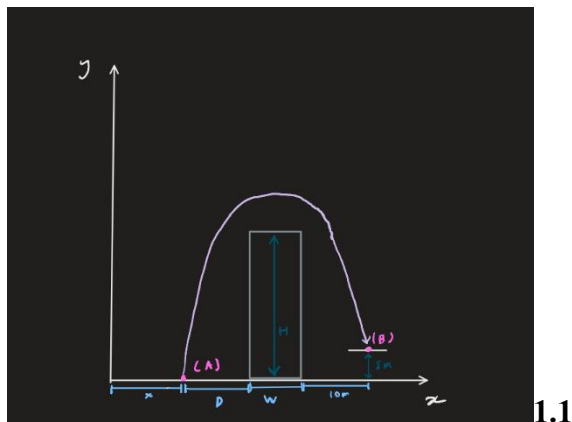
The axis we are using for this model:

We are considering the axis as shown in this figure 1.0:



- The motion will take effect on the x and y axis to eliminate any motion or displacement on the z axis to further simplify the model and the calculation.
- The launch point at A and landing at point B will all take place in the first quadrant of the positive x and y axis.
- The object (the stone) will be traveling in the direction of the positive x -axis.

The initial illustrating diagram of the model 1.1 :



- x is the horizontal distance from point A to the axis.
- D is the horizontal distance from point A to the building.

- L is the width of the building.
- H is the height of the building.
- A is the point of launching, and B is the point of landing.
- The point of landing B will be 10m away from the building on the x-axis and 5m away from the building on the y-axis.

2. Equations needed and dynamics used:

The mass of the object will not be considered in the projectile, the object will be launched at an initial speed (u) from a point (A) at a projection angle of (θ), the object will be in projectile motion given that the acceleration will be constant throughout the journey as the horizontal component of velocity will remain constant therefore the horizontal acceleration will be 0 and as the vertical acceleration is a constant due to the earth's gravitational field which is $g = 9.81ms^{-2}$, neglecting the drag, or as it is known as air resistance. Given that, we can assume that:

- The initial Horizontal velocity (U_h) = $u \cos \theta$
- The initial Vertical velocity (U_v) = $u \sin \theta$
- The Horizontal displacement from A (S_h) = $S_{h0} + U_h t$
- The Vertical displacement from A (S_v) = $S_{v0} + U_v t + \frac{1}{2} g t^2$
- Time taken to reach point B (t) = $\frac{D+L+10}{u \cos \theta}$
- Therefore, the equation of Vertical displacement becomes (S_v) = $U \sin \theta \times \frac{(D+L+10)}{U \cos \theta} + \frac{g}{2} \times \left(\frac{D+L+10}{U \cos \theta} \right)^2$
- The equation we obtain is $H - Y = \tan \theta \left(D + \frac{L}{2} \right) + \frac{g \left(D + \frac{L}{2} \right)^2}{2 U^2 \cos^2 \theta}$
- After we simplify, taking the peak point where vertical velocity is 0 hence the vertical component $u \sin \theta = 0$, therefore we obtain the equation $u = \sqrt{\frac{g \left(D + \frac{L}{2} \right)^2}{H - Y}}$
- Taking the displacement about point B, we obtain the equation $5 - Y = \tan \theta (D + L + 10) + \frac{g}{2} \left[\frac{(D+L+10)^2}{u^2 \cos^2 \theta} \right]$, we can substitute the u value obtained from the above equation and get the angle θ
- Another equation that can be used for determining different velocity components is $u = \sqrt{U_h^2 + U_v^2}$
- To get total distance we apply the Pythagoras theorem, $S_T = \sqrt{S_h^2 + S_v^2}$, this though would give us the modulus value and no direction

3. Matlab code development:

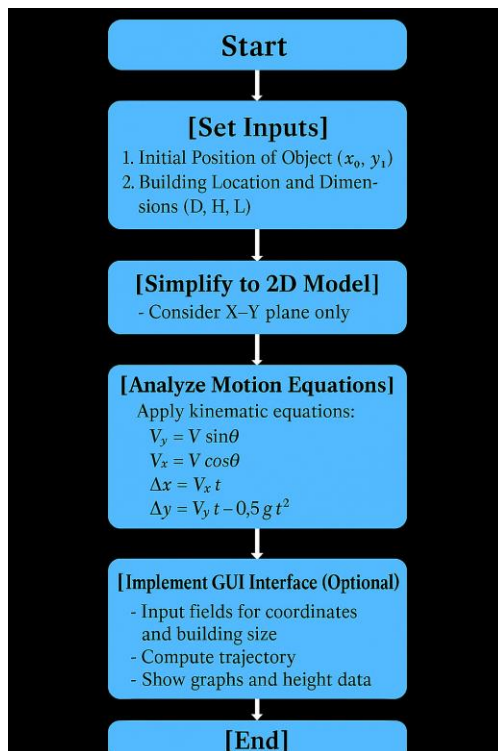
A MATLAB script is written to:

- Accept user-defined inputs:
 - Launch coordinates (x_1 , y_1)
- Building height (H) and width (L)
 - Target coordinates
- Compute:
 - Required launch angle and velocity
 - Time of flight
 - Maximum height
- Use plot() to graph the projectile trajectory.
- Include checks to determine whether the projectile passes above the building.

4. GUI implementation:

A user-friendly GUI is built using MATLAB App Designer or uifigure.

- allows switching between command-line and GUI modes.
- Inputs: launch point, target point, building dimensions.
- Outputs: graphical trajectory, numerical values of angle, velocity, time of flight, and result validation.



The Code:

```
% Get input parameters from user
disp('Enter the launch parameters:');
x0 = input('Initial x-position (m): ');
y0 = input('Initial y-position (m): ');
D = input('Distance to building (m): ');
H = input('Building height (m): ');
W = input('Building width (m): ');
% Fixed target position (10m beyond building, 5m above ground)
target_x = x0 + D + W + 10;
target_y = 5;
% Physics constants
g = 9.81; % gravity (m/s^2)
% Try different angles to find a solution
found_solution = false;
for angle = 1:89
    theta = angle; % Launch angle in degrees
    % Calculate required time to reach target_x
    delta_x = target_x - x0;
    v0_x = delta_x / (sqrt( 2*(delta_x * tand(theta) - (target_y - y0))) / g ));
    % Calculate required velocity components
    v0 = v0_x / cosd(theta);
    % Check if this clears the building (now checking entire width)
    time_to_building_start = D / (v0 * cosd(theta));
    time_to_building_end = (D + W) / (v0 * cosd(theta));
    % Check height throughout building width
    clear_building = true;
    for t_check = linspace(time_to_building_start, time_to_building_end, 20)
        y_at_check = y0 + v0 * sind(theta)*t_check - 0.5*g*t_check^2;
        if y_at_check <= H + 0.1 % Small clearance margin
            clear_building = false;
            break;
        end
    end
    if clear_building
        found_solution = true;
        break;
    end
end
% Display results
if found_solution
    fprintf('\nSolution found:\n');
    fprintf('Launch velocity: %.2f m/s\n', v0);
    fprintf('Launch angle: %.2f degrees\n', theta);
    % Calculate and plot trajectory
    flight_time = (delta_x) / (v0 * cosd(theta));
```

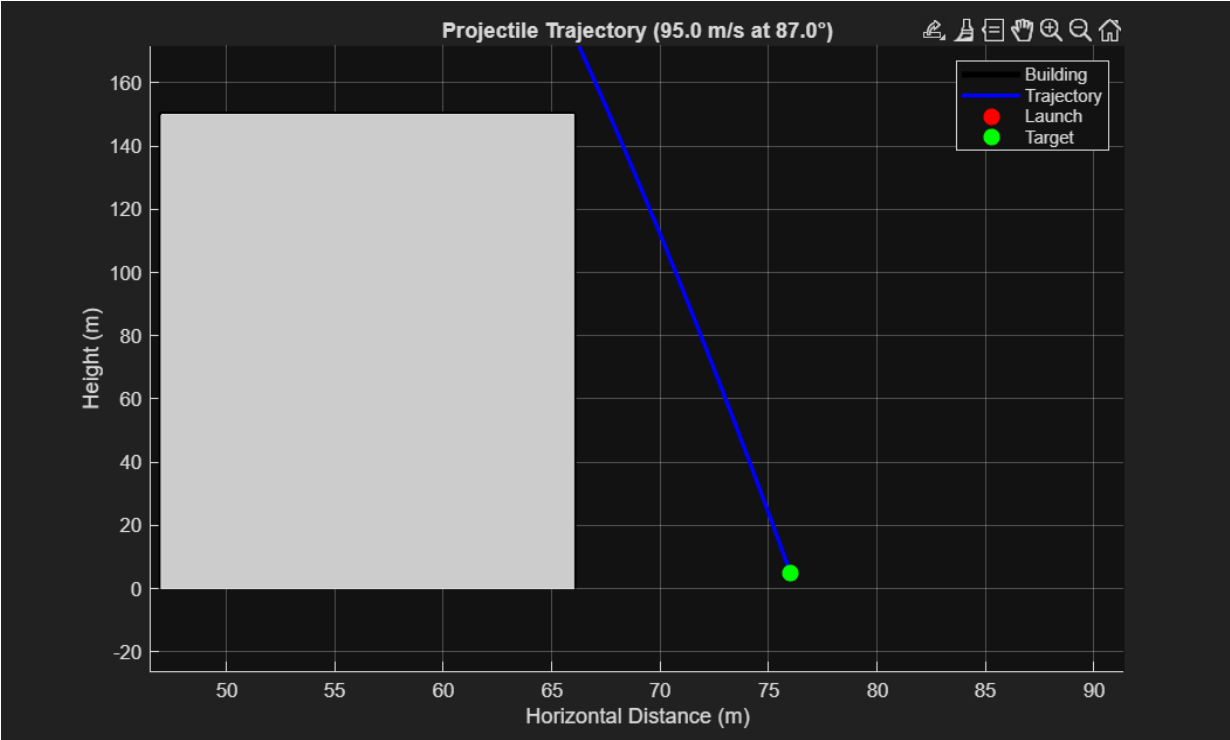
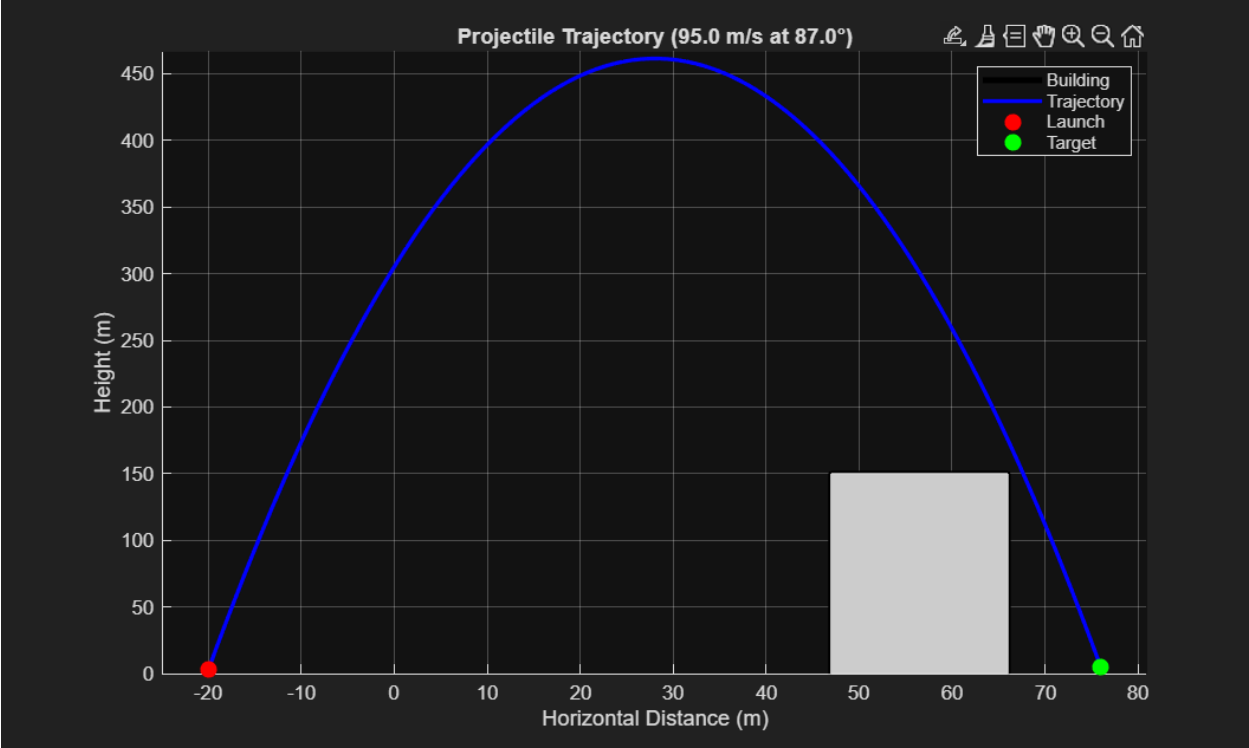
```

t = linspace(0, flight_time, 100);
x = x0 + v0 * cosd(theta) * t;
y = y0 + v0 * sind(theta) * t - 0.5 * g * t.^2;
% Create plot
figure;
hold on;
grid on;
% Plot building (now with width)
building_x = [x0+D, x0+D, x0+D+W, x0+D+W];
building_y = [0, H, H, 0];
plot(building_x, building_y, 'k-', 'LineWidth', 3);
fill(building_x, building_y, [0.8 0.8 0.8]);
% Plot trajectory
plot(x, y, 'b-', 'LineWidth', 2);
% Plot points
plot(x0, y0, 'ro', 'MarkerSize', 8, 'MarkerFaceColor', 'r');
plot(target_x, target_y, 'go', 'MarkerSize', 8, 'MarkerFaceColor', 'g');
% Labels
xlabel('Horizontal Distance (m)');
ylabel('Height (m)');
title(sprintf('Projectile Trajectory (%.1f m/s at %.1f°)', v0, theta));
legend('Building', '', 'Trajectory', 'Launch', 'Target');
% Set axis limits
xlim([min(x0, target_x)-5 max(x0, target_x)+5]);
ylim([0 max([y0, max(y), H, target_y])+5]);
else
fprintf('\nNo solution found with these parameters.\n');
fprintf('Try increasing the initial height or decreasing the building height/width.\n');
end
end

```

Notes:

1. The code begins by prompting the user to enter launch parameters
2. The target position is fixed at 10 meters beyond the building and 5 meters above ground
3. Gravitational acceleration is defined as a physical constant
4. The code tries launch angles from 1 to 89 in 1-degree increments
5. For each iteration, the code calculates the horizontal distance to target,
 - Then calculates the required horizontal velocity component using the projectile motion equation rearranged to solve for velocity
 - Finally it calculates the total initial velocity from the horizontal component and angle
 - It then calculates the time range when the projectile is directly above the building, and checks 20 points across this time range to ensure the projectile clear the entire width of the building,
 - Uses a small clearance margin of 0.1 meters above the building height
 - If any clearance check fails the angle is rejected
 - The loop breaks when the first valid solution is found or all angles are attempted and rejected
6. It then displays the found launch velocity and angle
7. Generates 100 points along the flight path
8. Creates a figure with a grid
9. Draws the building as a grey rectangle with a black outline
10. Plots the trajectory as a blue line
11. Marks the launch point as a red dot and the target as a green one
12. Adds axis labels, a title, and legend
13. Sets appropriate axis limits
14. If not solution is found, displays that to the user and suggests parameter adjustments



```
>> Porjectile_Project
Enter the launch parameters:
Initial x-position (m): -20
Initial y-position (m): 3
Distance to building (m): 67
Building height (m): 150
Building width (m): 19
```

```
Solution found:
Launch velocity: 94.97 m/s
Launch angle: 87.00 degrees
>> |
```

MATLAB App

Design View

Code View

Initial X

Initial Y

Distance to Building

Building Height

Building Width

Plot Projectile



Conclusion:

The purpose of this project was to analyse the projectile of a launch of an object from point A to point B, a Matlab simulation was created to give us the launch angle and the initial velocity if the object is launched from different coordinates(point A), mass was not taken into consideration and there was no movement in the z axis, the simulation also plots graphs of the projection as presented in The Code section.